Assignment

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Solution

This Python program asks for the user's name, keeps the conversation running with while True, cleans input using lower().strip(), recognizes greetings, answers multiple topics, uses if/elif/else, and exits when the user enters bye or exit.

```
name = input("What is your name? ").strip()

print(f"Hello {name}! I am a simple chatbot.")
print("You can ask me about Python, AI, or programming.")
print("Type 'bye' or 'exit' to end the conversation.")

while True:
    message = input(f"{name}: ").lower().strip()

    if "bye" in message or "exit" in message:
        print(f"Goodbye {name}! Have a great day!")
        break

    elif "hello" in message or "hi" in message:
        print(f"Hello {name}! How can I help you?")

    elif "python" in message:
        print("Python is a simple and powerful programming language.")

    elif "ai" in message or "artificial intelligence" in message:
        print("AI enables computers to perform tasks that normally require human intelligence.")

    elif "programming" in message:
        print("Programming is the process of writing instructions for a computer.")

    else:
        print("Sorry, I do not understand that. Try asking about Python, AI, or programming.")
```

How the Solution Meets the Requirements

Ask the user's name: input() is used before the main conversation.

Use while True: The while True loop keeps the chatbot running.

Use input(): input() receives each user message.

Clean input: lower().strip() normalizes the message.

Recognize greetings: The chatbot recognizes hello and hi.

Answer three topics: It responds to Python, AI, and programming.

Use if / elif / else: These statements control the chatbot's decisions.

Use break: bye or exit ends the loop.

Use the user's name: The name appears in greeting and exit responses.

Unknown message: else provides a useful fallback response.

Sample Run

```
What is your name? Sandeep
Hello Sandeep! I am a simple chatbot.
You can ask me about Python, AI, or programming.
Type 'bye' or 'exit' to end the conversation.

Sandeep: hello bot
Hello Sandeep! How can I help you?

Sandeep: what is python
Python is a simple and powerful programming language.

Sandeep: what is AI
AI enables computers to perform tasks that normally require human intelligence.

Sandeep: what is the weather
Sorry, I do not understand that. Try asking about Python, AI, or programming.

Sandeep: bye
Goodbye Sandeep! Have a great day!
```

Flow: **INPUT** → **PROCESS** → **RESPOND** → **REPEAT** → **EXIT**